11	**(a)**	**(i)**	Force = pressure × area $= rg\left(2x\right)A = 2Argx$	B1
		(ii)	mass of liquid $= rAL$	C1
			Hence $F = ma$ gives $2Argx = \left(rAL\right)a$	A1
			cancelling *A* and rearranging gives $a = \dfrac{2gx}{L}$	A0
		(iii)	When liquid is displaced *x* upwards, the restoring force is downwards.	B1
			L (and *g*) is constant	M1
			So *a* is directly proportional to *x*	A1
		(iv)	$w^2 = \dfrac{2g}{L}$,	C1
			and $w = \dfrac{2p}{T}$	C1
			$T = 2p\sqrt{\dfrac{L}{2g}} = 2p\sqrt{\dfrac{0.40}{2´\ 9.81}} = 0.90\,\text{s}$	A1
	(b)	**(i)**	displacement <u>decreasing with time</u>.	M1
			same period or slightly longer	A1
				M1
		(ii)	maximum speed $v_0 = wx_0 = \sqrt{\dfrac{2g}{L}}x_0 = \sqrt{\dfrac{2´\ 9.81}{0.40}}´\ 0.035 = 0.25\ \text{m s}^{-1}$	

		maximum kinetic energy, $K_0 = \frac{1}{2}mv_0^2 = \frac{1}{2} \times 0.50 \times 0.245^2 = 0.015\,\text{J}$	A1
	(iii)	1. shape is correct: negative quadratic graph (judge by eye)	B1
		axes are labelled correctly: K_0 found in (b)(ii) <u>and</u> maximum displacement at ± 3.5 cm	B1
		2.quadratic graph.	B1
(c)	**(i)**	$f = \frac{1}{T} = \frac{1}{0.897} = 1.1\text{ s}$ reciprocal of value calculated in (a)(iv)	B1
	(ii)	occurs when frequency of oscillator is close to / equal the natural frequency of oscillation of water column	B1
		energy transfer to oscillating water column is most efficient / maximum	B1
	(iii)	lower maximum amplitude and shift slightly to the left (allow the same f_0)	B1